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Executive Summary

- Seasoned technology executive with 19+ years driving the architecture and delivery of high-volume financial trading platforms. Specialized in **legacy-to-cloud migrations** and building secure, scalable ecosystems that fuel business growth and AI-driven innovation for leading global institutions.
 - Proven leader in designing complex, mission-critical solutions that process **billions in daily transactions**, consistently delivering enhanced system reliability, rapid deployment cycles, and rigorous compliance with global regulatory standards.
 - Experienced in translating complex business requirements into robust, future-ready architectures empowering organizations to unlock new revenue streams, operate with resilience in fast-moving markets, and accelerate digital transformation.
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Technical Expertise

- **Microservices, Cloud-Native & Domain-Driven Design:** Architected and migrated legacy FX monoliths to cloud-native, microservices platforms using AWS, Docker, and Kubernetes driving 99.97% system uptime, 80% faster release cycles, and seamless scaling for high-volume institutional clients.
 - **Enterprise Technology Stack:** Engineered secure trading platforms leveraging Java/Spring Boot, Python, and event-driven architectures (Kafka, Solace, GraphQL, RESTful APIs), enabling real-time data pipelines and complex algorithmic trade workflows that consistently handled \$1B+ daily volume across global markets.
 - **Performance & Low-Latency Systems:** Designed and optimized market data systems and algorithmic trading engines with FIX Protocol, multicast technologies, and KDB+ achieving sub-millisecond execution speed and 10-15% improvement in trade processing efficiency for mission-critical, high-frequency strategies.
 - **Security, Compliance & DevOps:** Embedded OWASP security principles, automated pipelines, and DevSecOps practices throughout the CI/CD lifecycle, resulting in 40% fewer vulnerabilities and zero audit findings for platforms subjected to MiFID II, Dodd-Frank, and multi-jurisdictional requirements.
 - **Data Infrastructure & Scalability:** Delivered robust data foundations using PostgreSQL, Oracle, MongoDB, and advanced cloud orchestration enabling reliable, high-speed analytics for risk management, compliance reporting, and sustainable P&L accuracy improvements up to 25%.
 - **Domain Impact:** Consistently shipped globally compliant FX and risk management platforms, opening new revenue channels and enabling secure market expansion into APAC and EMEA for tier-one financial institutions.
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Applied Artificial Intelligence (AI)

Built working systems across the control, evaluation, and defense layers of production AI.

- Constitutional governance SDK that evaluates every LLM output against versioned plain-English rules with an append-only audit log, and a pre-execution code validator that blocks SQL injection, unsafe commands, and secrets exposure via AST rewriting before generated code reaches a runtime.
- Full-stack fairness audit platform calculating demographic parity and disparate impact across protected groups for financial ML models, and a prompt injection defense system with four layered strategies and live effectiveness tracking across multiple LLM backends.

- Multi-agent code reviewer that runs security, performance, and quality checks against git diffs simultaneously, and a real-time trade risk monitor on Kafka with dead letter queue handling.

Research Outputs / Preprints: SSRN (DOI): <https://dx.doi.org/10.2139/ssrn.6384060>.

Portfolio Repository: <https://github.com/sushilduseja>

Professional Experience

Barclays | Vice President, Solutions Architect – Platform Modernization | Singapore March 2023 – March 2025

Architectural Impact & Business Value

- **Enterprise Platform Modernization:** Led £6M technology transformation initiative, modernizing legacy FX trading systems into data-centric, cloud-ready microservices platform, improving deployment velocity by 80% and system availability to 99.97%.
- **Scalable Trading Infrastructure Design:** Architected single-dealer FX platform serving 100+ institutional clients, engineering high-scalability infrastructure supporting \$1B+ daily trading volume and enabling future AI-driven analytics capabilities.
- **Real-Time Data & AI Analytics Enablement:** Designed real-time data pipeline architecture processing 10M+ market events daily, creating analytics foundation for machine learning applications while leading independent R&D in AI security and governance frameworks.
- **Embedded DevSecOps & Regulatory Compliance:** Integrated DevSecOps principles across CI/CD pipeline and design lifecycle, achieving 40% reduction in security vulnerabilities and ensuring MiFID II/Dodd-Frank regulatory compliance.
- **Pricing & Workflow Strategy Validation:** Drove technical strategy for pricing and trade workflow components, successfully validating solution effectiveness through 10+ pilot deployments with major institutional clients.

Innovation & Operational Excellence

- **Automation Programs:** Executed large-scale automation and workflow optimization initiatives, reducing manual intervention by 40% and streamlining high-frequency trading operations for greater efficiency.
- **Metrics & Reliability:** Developed monitoring and KPI frameworks to ensure robust user experience, system reliability, and audit-proof performance sustaining operational integrity through consistent incident-free delivery cycles.

Commerzbank | Vice President, Delivery Lead – Algo Trading Solutions | Singapore September 2017 – July 2022

Platform Leadership & Revenue Growth

- **Global Team Technical Leadership:** Provided technical leadership for 13-member global trading technology team, driving end-to-end design and delivery of high-performance Java-based algorithmic trading systems supporting \$500M+ daily volume.
- **Strategic Platform Initiatives:** Architected and delivered 4 major trading platform initiatives directly contributing to 15% revenue growth, focusing on scalable Java/Spring Boot backends and high-availability system integrations across global markets.
- **Market Expansion & Product Launch:** Led technical strategy for TFX T-Clear integration, enabling market entry in Japan and capturing \$10M+ annual revenue opportunity, defining complex system interfaces using Java and high-throughput messaging queues.

Performance Optimization & Risk Control

- **Market Data & Latency Improvement:** Improved market data processing throughput by 10% and execution speed by 15% through low-latency FIX protocol bridges and market data adapters, processing 100K+ orders daily with sub-millisecond latency.
- **Real-Time Risk Management:** Engineered real-time risk and position-keeping services, improving P&L accuracy by 25% and delivering consistent, high-quality data feeds supporting \$1B+ risk exposure across algorithmic strategies.

IRIS Software Inc. | Associate Project Manager | India | 2017 – 2017

- **Client:** Bank of Montreal - DTCC trade confirmation and settlement platform
- Led requirements, analysis, and implementation of platform enhancements for high-volume trade processing
- Reduced application memory consumption by 30% through GC log analysis and systematic code refactoring

Mahindra Comviva | Technical Manager | India | 2016 – 2017

- **Product:** PreTUPS - prepaid recharge distribution and mobile account management platform
- Directed product development, release management, and feature delivery across distributed teams
- Cut AWS cloud spending by 25% through strategic optimization and best practice implementation
- Improved application stability by 15% by identifying and resolving code inefficiencies

R Systems International | Associate Technical Architect | India | US | 2015 – 2016

- **Client:** DMA Inc. Tax Interchange for Retail - enterprise point-of-sale tax solution
- Rebuilt client relationship and trust through consistent delivery and transparent communication
- Established architectural standards, design patterns, and code review frameworks
- Introduced Test-Driven Development (TDD), achieving 40% reduction in production defects
- Diagnosed and resolved performance bottlenecks using JVisualVM profiling tools

JK Tech | Project Lead | India | UK | 2009 – 2015

- **Client:** Specsavers Optical Group - enterprise retail and supply chain systems
- Managed complete project lifecycles from initiation through post-implementation review
- Improved Agile delivery efficiency by 20% through process refinement and team coaching
- Promoted three times in six years: System Analyst → Technical Lead → Project Lead
- Recognized with "Star of the Iteration" awards (2012, 2013) and "Star of the Quarter" (June 2011)

Fiserv | Senior Software Engineer | India | US | 2007 – 2009

- **Client:** Fiserv Trade Order Management - institutional trading platform
- Designed and built automated test framework executing 500+ tests in 7 minutes, saving 40% processing time
- Developed proof-of-concept solutions using emerging technologies (OpenESB, Apache ServiceMix)
- Delivered specifications, program changes, and comprehensive technical documentation

JK Tech | Software Engineer | India | UK | 2005 – 2007

- **Internal Products:** iTimer timesheet application and corporate website modernization
 - Introduced automated unit testing frameworks, reducing manual testing effort by one man-day monthly
 - Mastered Agile and iterative development methodologies in production environments
 - Supported QA, UAT, and production stabilization across multiple concurrent projects
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Certifications & Executive Education

AI Strategy & Safety

AGI Strategy, BlueDot Impact 2026
Future of AI, BlueDot Impact 2026
Generative AI for Executives 2024

Cloud & Emerging Technology

Cloud-Native & Multicloud Apps 2022
Blockchain Essentials 2022

Cybersecurity & Governance

ISO/IEC 27001 Lead Auditor 2025
AI Security & Governance 2025
AI Security Fundamentals 2025
Cybersecurity GRC 2024
Foundations of Cybersecurity 2023

Community & Thought Leadership

AI Safety & Governance Community
Active contributor · 2023-Present

Key Differentiators

- **Strategic Vision:** Defines and delivers architectures that drive growth, resilience, and regulatory success for high-volume financial platforms.
- **Global Systems Integration:** Expert in architecting platforms for APAC/EMEA markets, navigating complex compliance and interoperability.
- **Innovation Leadership:** Pilots and embeds modern technologies - AI governance, DevSecOps, microservices - to future-proof enterprise systems.
- **Cross-Functional Influence:** Builds trust and alignment across technical and business teams, making architecture a catalyst for collaboration and delivery.